UI & UX in Human–Computer Interaction

Understanding interfaces, experiences, principles, process and real-world examples



Learning Objectives

- Define User Interface (UI) and User Experience (UX).
- Differentiate UI from UX and explain how they work together.
- Identify key principles of effective interface design.
- Describe the UX design process and common evaluation methods.
- Analyze familiar apps/websites using UI and UX concepts.

What is HCI?

- HCI studies how people interact with computing systems.
- Human: users, abilities, needs, goals, emotions and limitations.
- Computer: software, hardware, displays, sensors and interactive systems.
- Interaction: communication and actions between people and technology.

HCI = 3 Parts

Human
Computer
Interaction

What is User Interface (UI)?

- UI is the visual and interactive layer through which users control a digital product.
- Includes buttons, menus, icons, typography, colors, forms, navigation and layouts.
- Good UI makes actions understandable, consistent and visually clear.
- Example: A mobile banking app's Login button, PIN field, icons and navigation bar.

UI Example

Mobile banking screen

- Login
- PIN field
- Navigation
- Icons

What is User Experience (UX)?

- UX is the overall experience a person has while using a product or service.
- It includes usefulness, ease of use, accessibility, efficiency, trust and emotional response.
- UX considers the complete journey—not only the screen design.
- Example: Food ordering: finding food → selecting → payment → tracking → delivery.

UX Question

Can the user achieve the goal easily, quickly and confidently?

UI vs UX — Key Difference

- UI focuses on how the product looks and behaves: colors, typography, buttons, icons, layout and interaction states.
- UX focuses on how the product works and feels: user goals, navigation, task flow, accessibility and satisfaction.
- UI is one important part of UX; both should support the same user goal.

UI

Looks + controls + interaction

UX

Journey + usability + satisfaction

How UI and UX Work Together



Example: E-commerce

UX decides the shopping journey; UI presents it through product cards, filters, buttons and checkout screens.

Core UI Design Principles

Consistency

Use familiar patterns, spacing and terminology.

Visibility

Make important options and system status easy to see.

Feedback

Show users what happened after an action.

Simplicity

Avoid unnecessary information and controls.

Hierarchy

Use size, position and contrast to show importance.

Accessibility

Design for users with different abilities and contexts.

UX Design Process



Example

Interview students → define attendance problems → prototype screens → test → refine.

UX Design Process

1. Empathize — understand users
2. Define — identify the problem
3. Ideate — generate solutions
4. Prototype — build a testable version
5. Test — observe users
6. Iterate — improve based on evidence

The process is iterative, not strictly one-time.

Research Goal

Understand real user needs.

User-Centered Design (UCD)

Design decisions start with users' needs, goals, abilities, and context.

Users are involved through research, testing, and feedback.

Design should solve a real problem—not just look attractive.

Ask: Who are the users? What are they trying to achieve? What prevents success?

- Interviews: discover goals, problems and expectations.
- Observation: watch users perform real tasks.
- Surveys: collect patterns from many users.
- Analytics: measure time, errors, completion and drop-o

Research Goal

Understand real user needs.

Understanding Users

Research methods: interviews, surveys, observation, contextual inquiry, analytics, and usability testing.

Qualitative data explains why/how; quantitative data shows how many/how often.

Look for goals, pain points, behaviors, motivations, skills, and constraints.

Do not assume the designer is the same as the user.

Research Goal

Understand real user needs.

User Research in UX

- Interviews: discover goals, problems and expectations.
- Observation: watch users perform real tasks.
- Surveys: collect patterns from many users.
- Analytics: measure time, errors, completion and drop-off.

Research Goal

Understand real user needs.

Usability — Can Users Complete the Task?

- Learnability: Can a new user understand it quickly?
- Efficiency: Can users complete tasks with reasonable effort?
- Memorability: Can users return later and remember how it works?
- Errors: Does the design prevent errors and support recovery?
- Satisfaction: Does the experience feel comfortab

Test Task

"Find your last exam result." Measure time, errors and success.

User Journey Map

Shows the stages a user passes through while completing a goal.

Typical fields: stage, action, thoughts, emotions, pain points, and opportunities.

Example: University portal → Login → Find course → Download material → Submit assignment → Confirmation.

Journey maps reveal where UX can improve?

Activity: Design User Journey Map of your univer
…..!!!!!!!!

Test Task

"Find your last exam result." Measure time, errors and success.

Gestalt Principles in HCI

Gestalt principles explain **how people see and understand things on a screen**.

In **HCI (Human-Computer Interaction)**, designers use these principles to make websites and apps **easy to understand and use**.

1. Proximity (Near Things = Related)

Things placed **close together** look like they belong together.

Example:

📧 Email

🔒 Password

If these fields are close together, users understand they are part of the **login form**.

2. Similarity (Similar Things = Same Group)

Things that look **similar** are considered related.

Example:

All blue buttons → Similar actions

All red buttons → Delete/Warning actions

🔵 Save 🔵 Edit 🔵 Submit

---

3. Closure (Our Brain Completes Shapes)

Our brain can recognize an **incomplete shape** as complete.

Example:

Even if a circle has some missing lines, we can still see it as a **circle**. ⭕

4. Continuity (We Follow Lines and Patterns)

Our eyes naturally follow **lines, paths, and directions**.

Example:

➡️ Step 1 → Step 2 → Step 3 → Step 4

This helps users understand a process.

---

5. Figure-Ground (Main Object vs Background)

We separate the **important object (figure)** from the **background (ground)**.

Example:

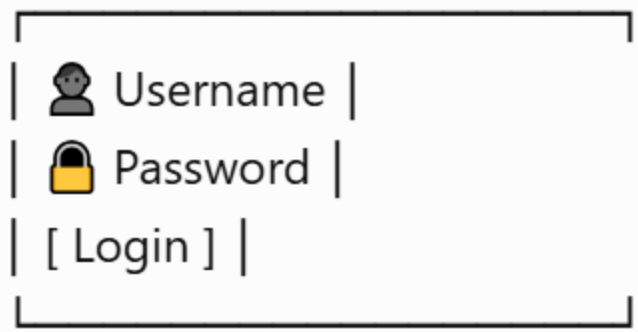
🖼️ A popup window appears clearly in front of a dark background.

The user immediately knows what to focus on.

6. Common Region

Items inside the **same box or area** are seen as one group.

Example:



Users understand these items belong together.

Visual Design: Gestalt Principles

Class Activity: Discover Gestalt Principles Yourself 🎯
Activity Type: Group Activity
Time: 20 minutes
Topic: Gestalt Principles in HCI
Objective

Students will **observe examples themselves** and identify how Gestalt principles work instead of only memorizing definitions.

Visual Design: Gestalt Principles

Step 1: Make Groups 👥
Give each group **one Gestalt principle**:
Group 1 → **Proximity**
Group 2 → **Similarity**
Group 3 → **Closure**
Group 4 → **Continuity**
Group 5 → **Figure-Ground**
Group 6 → **Common Region**

1. What do you think this principle means?
2. Find a real-life example.
Examples can be from:
•Mobile apps 📱
•Websites 💻
•Classroom objects 🏫
•Social media
•ATM screens
•Google or WhatsApp

Information Architecture & Navigation

- Information Architecture organizes content so users can find what they need.
- Use meaningful categories, labels and navigation structures.
- Keep navigation predictable across screens.
- Example: University portal → Dashboard → Courses → Attendance → Course Details.

Accessibility in UI/UX

- Use sufficient color contrast and readable text.
- Do not communicate information using color alone.
- Provide labels and meaningful error messages.
- Support keyboard navigation and assistive technologies.
- Use clear language and adequately sized touch targets.

Common UI/UX Problems — Before vs After

- BEFORE: too many fields, unclear labels, weak hierarchy, late error messages, hidden primary action.
- AFTER: grouped fields, clear labels, strong hierarchy, inline validation and one obvious primary action.

Real-World Examples

- Google Search: minimal UI and fast task completion.
- ATM: safe, understandable steps with clear prompts and confirmation.
- Food Delivery App: search, filters, cart, payment and tracking form a complete journey.
- University LMS: navigation, courses, assignments and grades should be easy to find.

UI/UX Evaluation Methods

- Usability testing: give users realistic tasks and observe difficulties.
- Heuristic evaluation: experts inspect an interface using usability principles.
- A/B testing: compare two design versions using measurable outcomes.

UI/UX and HCI: Putting It All Together

- HCI provides the broader study of people, technology and interaction.
- UX focuses on the complete user experience and journey.
- UI focuses on the interface users see and interact with.
- Usability, accessibility, consistency and feedback improve interaction quality.
- The best design starts with real user needs—not only visual appearance.

Class Activity: UI/UX Critique

- Choose a familiar app or website.
- Identify 3 UI elements: buttons, icons, colors, typography or layout.
- Identify 3 UX factors: navigation, errors, satisfaction or task completion.
- Find one usability problem and propose a better design.
- Present findings in 3–5 minutes.

Key Takeaway

Good UI makes interaction clear.

Good UX makes the whole experience useful, easy and satisfying.

Questions?